

Lecture 17-20

Chapter 4: Bivariate and Multivariate Random Variable

Math 4441: Probability and Statistics

Reference: Goodman & Yates – Introduction to Probability and Stochastic Process, 3rd Edition

Introduction

- Chapter 2 and Chapter 3 explained random experiments with outcome as a single numerical value from a discrete and continuous set.
- The probability models were developed for a discrete and a continuous random variable
- The probability models include
 - Probability mass function (PMF)
 - Probability density function (PDF)
 - Cumulative distribution function (CDF)

- In this Chapter, we will study random experiments with two numerical values as outcomes
 - Both from discrete set (Bivariate discrete distribution)
 - Both from continuous set (Bivariate continuous distribution)
 - One from a discrete set and the other from a continuous set (Bivariate mixed distribution)
- For example, in the classification of a set transmitted and received signals, each signal can be classified as – high, medium and low quality
- Random values might represent
 - Number of high-quality signals received
 - Number of low-quality signals received
- Accordingly, we can use random variables X and Y to represent the random values
- The simultaneous randomness of X and Y or the joint probability distribution of them will represent the probability model

- Similarly, two random values from two continuous sets might be associated with a single random experiment,
 - Need to probability model for simultaneous behavior of two continuous random variables, X and Y .
- Finally, a bivariate mixed distribution deals with probability models associated with one discrete and one continuous random variables

Bivariate discrete distributions

A bivariate discrete distribution is associated for random experiments with

- A pair of discrete random variables
- Joint discrete random variables
- Bivariate discrete random variables

And the probability models include:

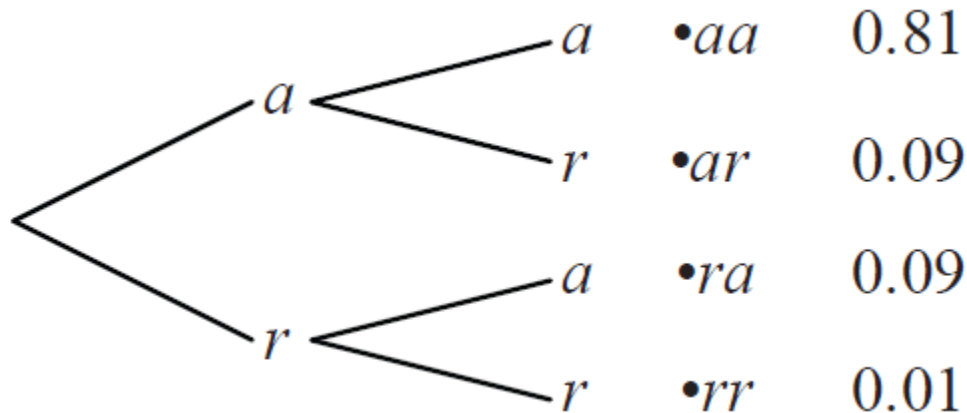
- Joint probability mass function (JMPF)
 - Marginal probability mass function (MPMF)
- Joint cumulative distribution function (JCDF)
 - Marginal cumulative distribution function (MCDF)

Example: Test two integrated circuits one after another. On each test, the possible outcomes are **a (accept)** and **r (reject)**. Assume that all circuits are acceptable with probability 0.9 and that the outcomes of successive tests are independent.

Count the number of acceptable circuits and the number of acceptable circuits before the first reject. (If both tests are successful, # of accepted circuits before the first reject is 2)

Develop the probability models for the experiment.

The experiment has the following tree diagram



The sample space of the experiment

$$S = \{aa, ar, ra, rr\}.$$

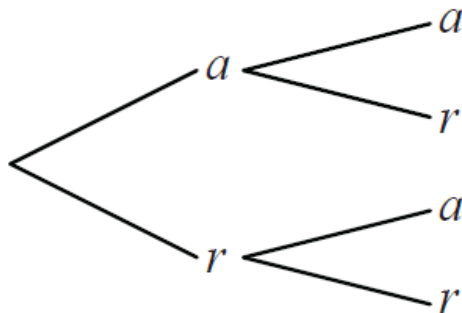
Random variables:

X : # of successful tests

$$S_X = \{0, 1, 2\}$$

Y : # of successful tests before the first reject

$$S_Y = \{0, 1, 2\}$$

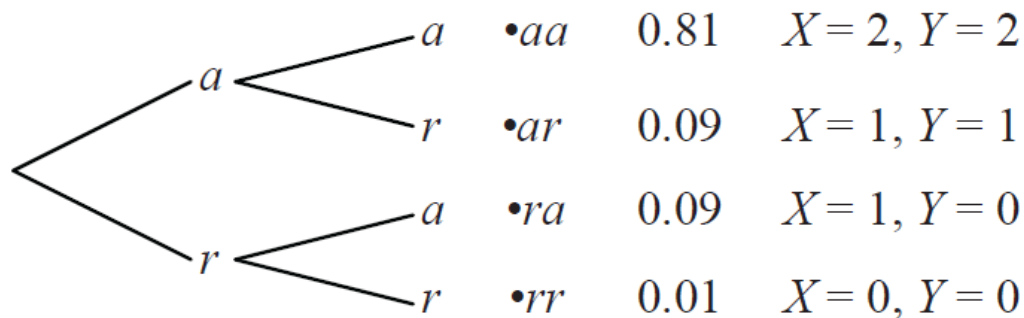


- Each outcome specifies a pair of values of X and Y .
- Single random variable: an outcome represents a point on the number line
- Bivariate random variables: an outcome represents a point on the 2-D plane
- Let $g(s)$ be the function that transform each outcome into a pair of random values. Then

$$g(aa) = (2, 2), \quad g(ar) = (1, 1), \quad g(ra) = (1, 0), \quad g(rr) = (0, 0)$$

- Observing the tree diagram, we have

$$P[aa] = 0.81, \quad P[ar] = P[ra] = 0.09, \quad P[rr] = 0.01$$



The probability model of a single discrete random variable represents

- Set of possible values (each value is associated with an event)
- It defines the probability of occurring each of the values as outcomes

The probability model of a pair of random variables represents

- Each pair of values as a point on a 2-D plane
- Set of all points on the 2-D plane
- Probabilities of occurring each point as the outcome of the experiment

The set of pairs of simultaneous values of X and Y ,

a point (x, y) on the 2-D plane,

Can be given by

$$S_{XY} = \{ (2, 2), (1, 1), (1, 0), (0, 0) \}$$

The probabilities of the pairs of values (or the points) as the outcomes are:

$$P[2, 2] = .91$$

$$P[1, 1] = 0.09$$

$$P[1, 0] = 0.09$$

$$P[0, 0] = 0.01$$

Joint Probability Mass Function (JPMF)

Definition 4.1 (Joint Probability Mass Function). The joint probability mass function of two discrete random variables X and Y , denoted as $P_{XY}(x, y)$, is defined as

$$P_{XY}(x, y) = P[X = x, Y = y],$$

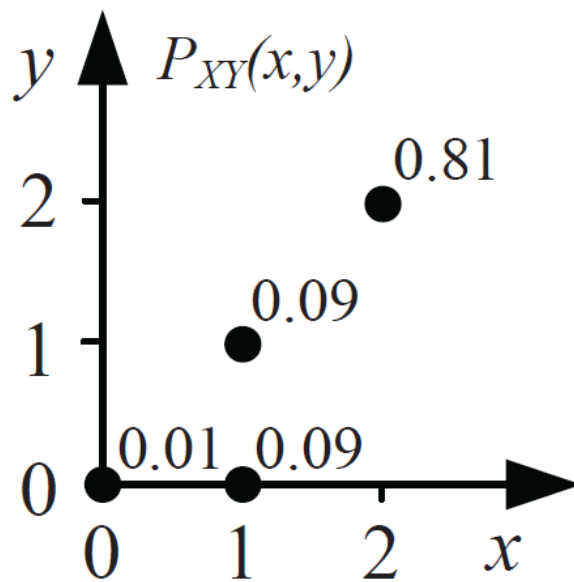
and it satisfies,

$$\begin{aligned} P_{XY}(x, y) &\geq 0; \\ \sum_x \sum_y P_{XY}(x, y) &= 1 \end{aligned}$$

The PMF of a single discrete random variable is represented as a list,

The JPMF of a pair of random variables can be represented in a number of ways.

The bivariate PMF or the JPMF can be represented as a graph



The bivariate PMF or JPMF can be represented as a list as well

$$P_X(x) = \begin{cases} 0.81, & x = 2, y = 2 \\ 0.09, & x = 1, y = 1 \\ 0.09, & x = 1, y = 0 \\ 0.01, & x = 0, y = 0 \\ 0, & \text{otherwise} \end{cases}$$

A third presentation of the joint PMF is the matrix given below:

$P_{XY}(x, y)$	$y = 0$	$y = 1$	$y = 2$
$x = 0$	0.01	0	0
$x = 1$	0.09	0.09	0
$x = 2$	0	0	0.81

Marginal Probability Mass Function

Definition 4.2. If X and Y are two discrete random variables with joint probability mass function $P_{XY}(x, y)$, then the marginal probability mass function of X and Y are given by

$$\begin{aligned} P_X(x) &= P[X = x] \\ &= \sum_y P_{XY}(x, y), \end{aligned}$$

and,

$$\begin{aligned} P_Y(y) &= P[Y = y] \\ &= \sum_x P_{XY}(x, y). \end{aligned}$$

$$P_X(2) =$$

$$P_X(1) =$$

$$P_X(0) =$$

$P_{XY}(x, y)$	$y = 0$	$y = 1$	$y = 2$
$x = 0$	0.01	0	0
$x = 1$	0.09	0.09	0
$x = 2$	0	0	0.81

Similarly, the marginal PMF of Y is

$$P_Y(0) = \sum_{x=0}^2 P_{XY}(x, 0) = P_{XY}(0, 0) + P_{XY}(1, 0) = 0.10$$

$$P_Y(1) = \sum_{x=0}^2 P_{XY}(x, 1) = P_{XY}(1, 1) = 0.09$$

$$P_Y(2) = \sum_{x=0}^2 P_{XY}(x, 2) = P_{XY}(2, 2) = 0.81$$

and, $P_Y(y) = 0$, for $x \neq 0, 1, 2$.

Summary of the last class:

1. Bivariate distributions are used for experiments with two outcomes
2. Random variable X represents one value along the x -axis and Y represents the other value along the y -axis
3. Outcomes are an ordered-pair (x, y) and represents a point on the 2-D plane
4. JPMF of X and Y is defined as
$$P_{XY}(x, y) = P[X = x, Y = y]$$
And represents the probability of simultaneously occurring the value of x and y , i.e., occurring the even $\{X = x \cap Y = y\}$
5. Marginal PMFs of X and Y are given by

Expected Value and Variance from JPMF

Definition 4.3. If the joint PMF of X and Y is $P_{XY}(x, y)$, then $E[X]$ and $E[Y]$ are calculated as

$$\begin{aligned} E[X] &= \sum_x x P_X(x) \\ &= \sum_x x \left[\sum_y P_{XY}(x, y) \right] \\ &= \sum_x \sum_y x P_{XY}(x, y) \end{aligned}$$

Similarly,

$$E[Y] = \sum_y \sum_x y P_{XY}(x, y).$$

Definition 4.4. If the joint PMF of X and Y is $P_{XY}(x, y)$, then $Var[X]$ calculated as

$$\begin{aligned} Var[X] &= \sum_x (x - \mu_x)^2 P_X(x) \\ &= \sum_x (x - \mu_x)^2 \left[\sum_y P_{XY}(x, y) \right] \\ &= \sum_x \sum_y (x - \mu_x)^2 P_{XY}(x, y). \end{aligned}$$

Similarly,

$$Var[Y] = \sum_y \sum_x (y - \mu_y)^2 P_{XY}(x, y)$$

Joint Cumulative Distribution Function

Definition 4.5 (JCDF). The joint cumulative distribution function (JCDF) of random variables X and Y , denoted as $F_{XY}(x, y)$, is defined as

$$F_{XY}(x, y) = P[X \leq x \cap Y \leq y] = P[X \leq x, Y \leq y]$$

for all x and y . It is the probability of the intersection of two events: random variables X and Y therefore induce a probability distribution over a two-dimensional Euclidean plane.

$F_{XY}(x, y)$ has the following properties:

1. $F_{XY}(-\infty, -\infty) = F_{XY}(x, -\infty) = F_{XY}(-\infty, y) = 0$.
2. $F_{XY}(+\infty, +\infty) = 1$.
- 3.

$$F_{XY}(x, y) = \sum_{i=-\infty}^x \sum_{j=-\infty}^y P_{XY}(i, j).$$

any x_1, x_2, y_1 and y_2 such that $x_1 < x_2$ and $y_1 < y_2$, the probability $P[x_1 < X \leq x_2, y_1 < Y \leq y_2]$ is given in terms of $F_{XY}(x, y)$ by

$$P[x_1 < X \leq x_2, y_1 < Y \leq y_2] = F_{XY}(x_2, y_2) - F_{XY}(x_1, y_2) - F_{XY}(x_2, y_1) + F_{XY}(x_1, y_1).$$

How to represent the JCDF

Single Random Variable:

The CDF $F_X(x)$ represents –

The sum of the PMFs of all points on the number line that are

1. On the line $X = x$, or
2. To the left of the line $X = x$

$P_{XY}(x, y)$	$y = 0$	$y = 1$	$y = 2$
$x = 0$	0.01	0	0
$x = 1$	0.09	0.09	0
$x = 2$	0	0	0.81

Bivariate Random Variable:

The JCDF $F_{XY}(x, y)$ represents –

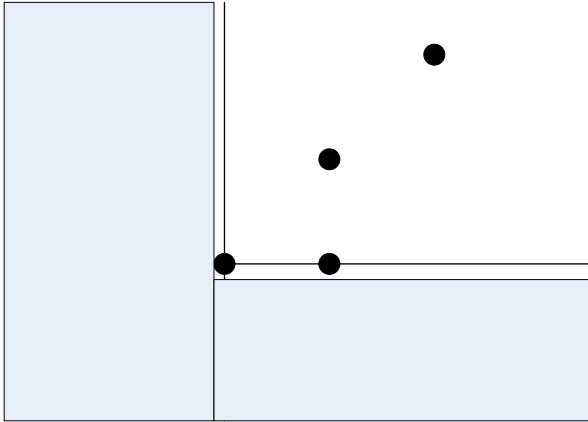
The sum of the JPMFs of all points on the 2-D plane that satisfy following two conditions:

1.
 - a. On the line $X = x$, or
 - b. To the left of the line $X = x$
2.
 - a. On the line $Y = y$, or
 - b. Below the line $Y = y$

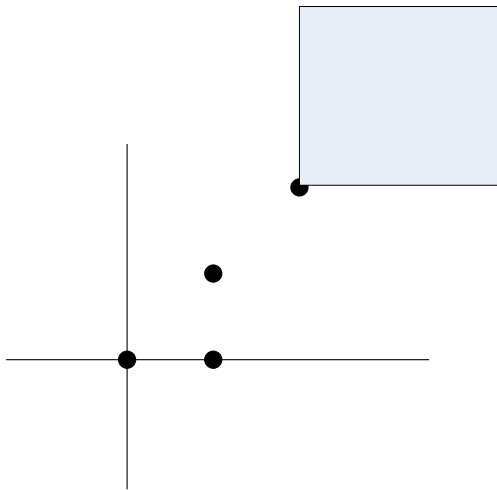
JCDF for random variable X and Y with PDF given by (Example 1)

$$P_X(x) = \begin{cases} 0.81, & x = 2, y = 2 \\ 0.09, & x = 1, y = 1 \\ 0.09, & x = 1, y = 0 \\ 0.01, & x = 0, y = 0 \\ 0, & \text{otherwise} \end{cases}$$

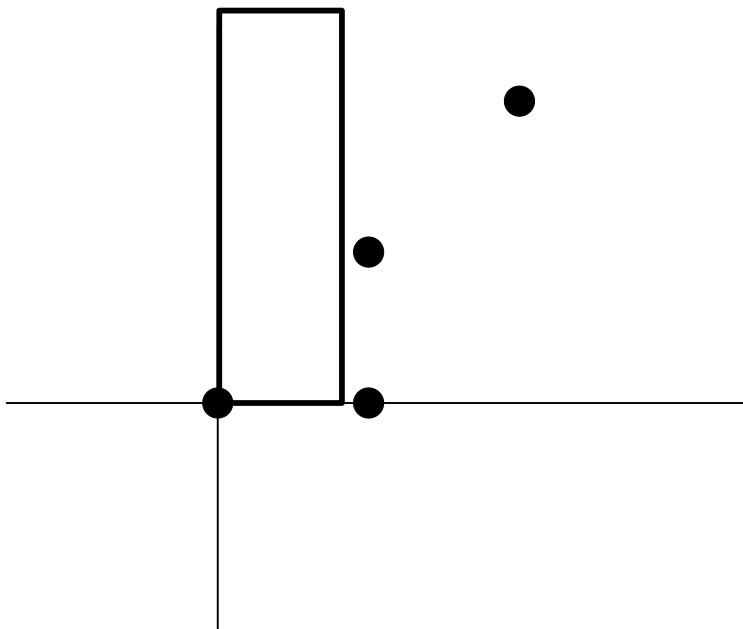
1. For $X < 0$ OR $Y < 0$



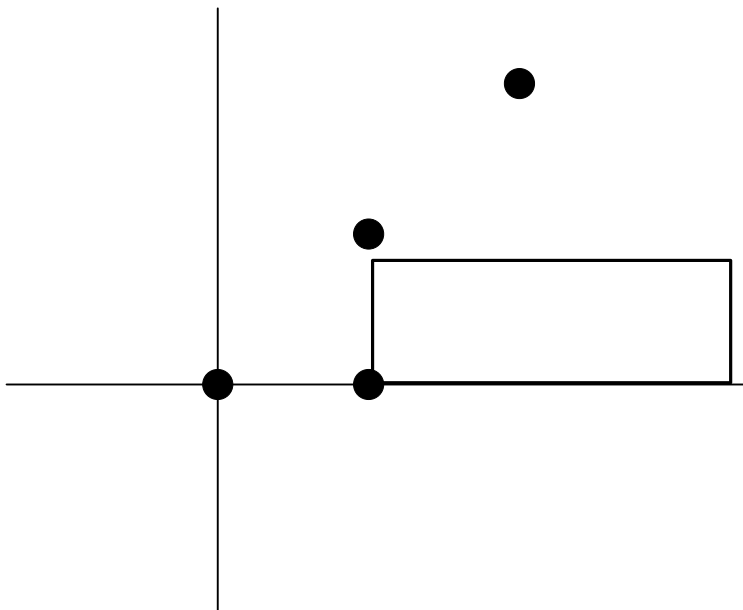
2. For $X \geq 2$ AND $Y \geq 2$



3. For $0 \leq X < 1$ AND $Y \geq 0$

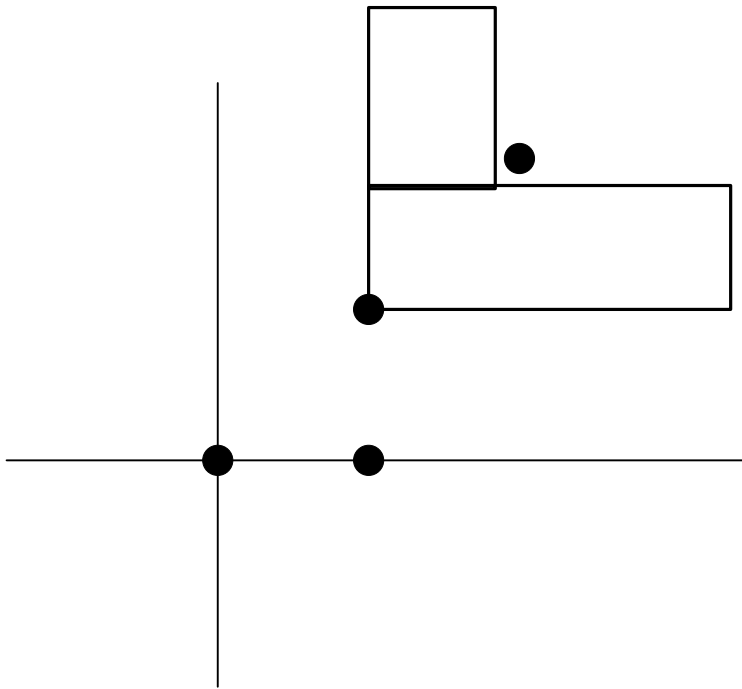


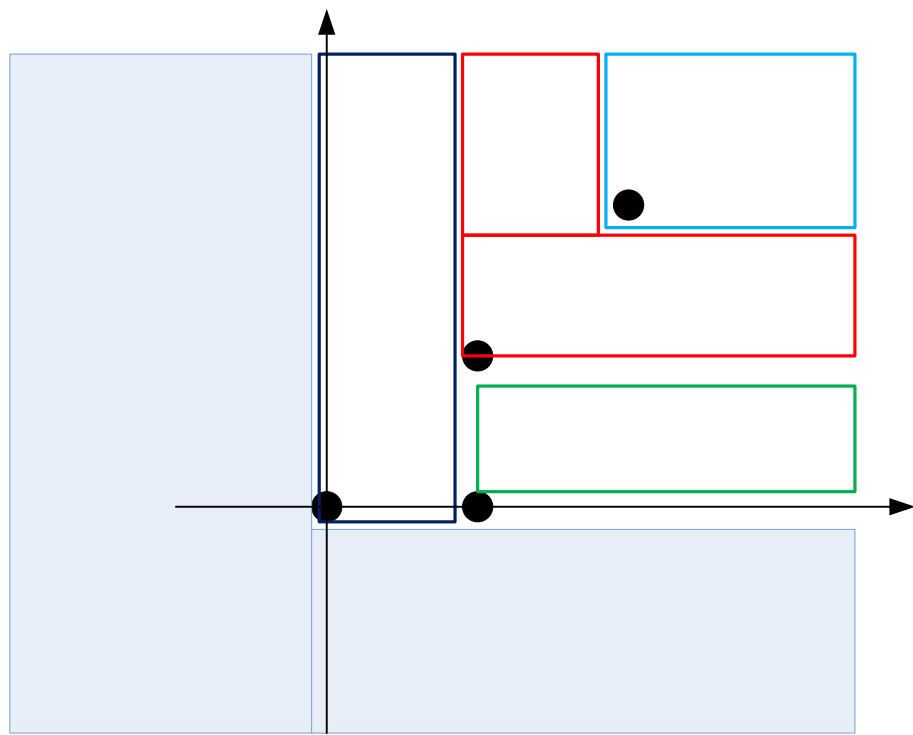
4. For $X \geq 1$ AND $0 \leq Y < 1$



5. For $X \geq 1$ AND $1 \leq Y < 2$

6. For $1 \leq X < 2$ AND $Y \geq 1$





Marginal Cumulative Distribution Function

Definition 4.6. If X and Y are two discrete random variables with joint CDF $F_{XY}(x, y)$, then marginal CDF of X and Y are

$$\begin{aligned} F_X(x) &= P[X \leq x] \\ &= P[X \leq x \cap Y \leq +\infty] \\ &= F_{XY}(x, \infty). \end{aligned}$$

The above relation follows from the fact that the joint event $X \leq x \cap Y \leq \infty$ is the same as the event $X \leq x$, since $Y \leq \infty$ is a **sure event**. Similarly,

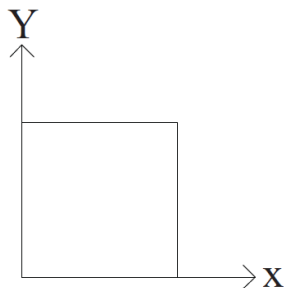
$$F_Y(y) = F_{XY}(\infty, y).$$

Bivariate Continuous Random Variables

1. Like a single continuous random variable PMF is meaningless
 - a. JPMF for 2 continuous RVs are not used
2. JCDF for bivariate continuous RVs are used
3. JPDP for bivariate continuous RVs are also used
4. Marginal PDF and Marginal CDF are used as well

Joint Cumulative Distribution Function (JCDF)

Consider 2 continuous random variables X and Y jointly continuous within a unit square



$$P[X = x, Y = y] = \frac{1}{n} \bigg|_{n=\infty} = 0.$$

define the interval for X as $[x_1, x_2]$ or $[x_1 \leq X \leq x_2]$

for Y as $[y_1, y_2]$ or $[y_1 \leq Y \leq y_2]$

the region $R = [x_1 \leq X \leq x_2, y_1 \leq Y \leq y_2]$

When $x_1 = 0$ and $x_2 = 1$ and $y_1 = 0$ and $y_2 = 1$.

R is equal the area of the square, which is 1.

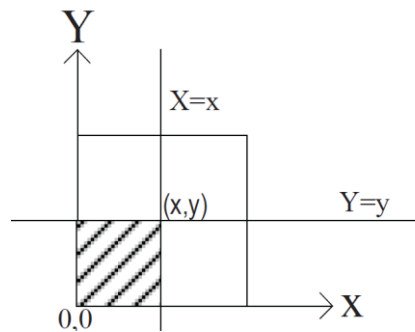
$$P[0 \leq X \leq 1, 0 \leq Y \leq 1] = \frac{\text{Area of the desired region } R}{\text{Total Area of the square}} = 1$$

$$x_1 = -\infty,$$

$$y_1 = -\infty$$

$$x_2 = x,$$

$$y_2 = y$$



$$P[-\infty \leq X \leq x, -\infty \leq Y \leq y] = P[X \leq x, Y \leq y].$$

Definition 4.7. The joint cumulative distribution function (JCDF) of a pair of continuous random variables X and Y , defined on a single sample space, is denoted by $F_{XY}(x, y)$, and defined by

$$F_{XY}(x, y) = P[X \leq x, Y \leq y],$$

and it represents the joint probability of occurring two events $\{X \leq x\}$ and $\{Y \leq y\}$ simultaneously.

Mathematical Representation of the CDF

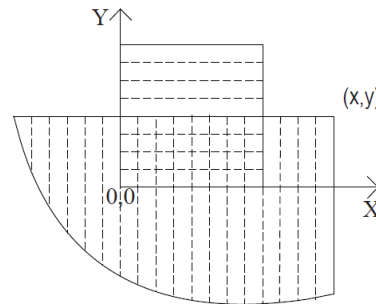
1. The joint CDF of a pair of continuous RVs has to be defined for all points on the $X - Y$ Plane.
2. The whole needs to be divided into sub-regions such that for each sub-region, a single function can define the CDF

How to calculate the JCDF for a pair of continuous RVs

1. Let X and Y are uniformly distributed within a unit square
2. CDF of the point (x, y) , $F_{XY}(x, y)$ is to be calculated
3. Let R denote the area for which X and Y are defined

$$R = [0 \leq X \leq 1, 0 \leq Y \leq 1]$$

4. JCDF for (x, y) defines the probability for a region, say R' ,
 $R' = [-\infty < X \leq x, -\infty < Y \leq y]$



5. JCDF defines the probability for the intersection of two regions, $R \cap R'$

1. Whenever either of the random variables is less than zero, the intersection of the area by the regions $[-\infty < X \leq x, -\infty < Y \leq y]$ and $[0 \leq X \leq 1, 0 \leq Y \leq 1]$ is empty. Therefore, if $X < 0$ or $Y < 0$ then the JCDF is zero, i.e.,

$$F_{X,Y}(x,y) = 0, \quad \text{if } x < 0 \quad \text{or} \quad y < 0.$$

2

The JCDF of any point where $x > 1$ and $0 \leq y < 1$ will include the double shared region with area $r_2 = y$ and the JCDF is given by

$$F_{X,Y}(x,y) = y \quad \text{if} \quad x > 1 \quad \text{or} \quad 0 \leq y < 1.$$

3

Any point within the shaded region will have a JCDF given by

$$F_{X,Y}(x,y) = x \quad \text{if} \quad y > 1 \quad \text{or} \quad 0 \leq x < 1.$$

The JCDF of any point (x, y) within the unit square can be represented by a single function. The JCDF is the area of the intersection of the unit square $[0 \leq x \leq 1, 0 \leq y \leq 1]$ and area covered by $F_{X,Y}(x, y)$ which is $[-\infty < X \leq x, -\infty \leq Y \leq y]$, which is $r_3 = x_4$ and the JCDF is given by

$$F_{X,Y}(x, y) = xy \quad \text{if} \quad 0 \leq x \leq 1, \quad 0 \leq y \leq 1.$$

The JCDF of any point within the shaded region can be represented by a single function $r_5 = 1$ and the JCDF is given by

$$F_{X,Y}(x,y) = 1 \quad \text{if} \quad x > 1, \quad y > 1.$$

Finally, the complete JCDF is given by

$$F_{X,Y}(x,y) = \begin{cases} 0, & \text{if } x < 0 \text{ or } y < 0; \\ y, & \text{if } x > 1, 0 \leq y \leq 1; \\ x, & \text{if } 0 \leq x \leq 1, y > 1; \\ xy, & \text{if } 0 \leq x \leq 1, 0 \leq y \leq 1; \\ 1, & \text{if } x > 1, y > 1 \end{cases}$$

Properties of JCDF

- i) The joint cumulative distribution function is a measure of probability, it must take on a value between 0 and 1, i.e.,

$$0 \leq F_{X,Y}(x, y) \leq 1$$

- ii) The random variables X and Y must assume real value. It is impossible for either of these two to take a value less than or equal to $-\infty$, and both must be less than $+\infty$. Therefore,

$$F_{X,Y}(-\infty, y) = 0$$

$$F_{X,Y}(x, -\infty) = 0$$

$$F_{X,Y}(-\infty, -\infty) = 0$$

$$F_{X,Y}(+\infty, +\infty) = 1$$

iii) The probability that X and Y have values within certain intervals can be calculated from their JCDF as shown in Fig. -.- and is given by

$$F_{X,Y}(x_1 \leq X < x_2, y_1 \leq Y < y_2) = F_{X,Y}(x_2, y_2) - F_{X,Y}(x_1, y_2) - F_{X,Y}(x_2, y_1) + F_{X,Y}(x_1, y_1).$$

Joint Probability Density Function

Definition 4.8. *Joint Probability Density Function.* The joint probability density function (JPDF) of jointly distributed random variables X and Y at point (x, y) is denoted as $f_{X,Y}(x, y)$ and is given by

$$\begin{aligned} f_{X,Y}(x, y) &= \lim_{\substack{\Delta x \rightarrow 0 \\ \Delta y \rightarrow 0}} \frac{\frac{\partial}{\partial x} F_{X,Y}(y + \Delta y) - \frac{\partial}{\partial x} F_{X,Y}(y)}{\Delta x \Delta y} \\ &= \frac{\partial^2}{\partial x \partial y} F_{X,Y}(x, y), \end{aligned}$$

$$f_{X,Y}(x, y) = \begin{cases} 1, & \text{for } 0 < x < 1, 0 < y < 1; \\ 0, & \text{otherwise.} \end{cases}$$

Properties of JPDF

- i. Since the JPDF represents a probability (i.e., probability per unit area), it must be nonnegative.

$$f_{X,Y}(x, y) \geq 0.$$

- ii. Since the JPDF represents probability per unit area, the JPDF multiplied by the total area will give the total probability which is 1.

$$\int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} f_{X,Y}(x, y) dy dx = 1.$$

- iii. An extension of the above property

$$F_{X,Y}(x, y) = \int_{-\infty}^x \int_{-\infty}^y f_{X,Y}(x, y) dy dx$$

iv. The probability of getting the pair of values in any bounded region is

$$P[x_1 < X \leq x_2, y_1 < Y \leq y_2] = \int_{x_1}^{x_2} \int_{y_1}^{y_2} f_{X,Y}(x, y) dy dx$$

v. If the two random variables are jointly uniformly distributed over the whole area associated with the random variables, then the JPDF is a constant.

$$f_{X,Y}(x, y) = c.$$

Example 4.6. Random variable X and Y have the joint PDF given by

$$f_{X,Y}(x,y) = \begin{cases} c, & \text{for } 0 \leq x \leq 5, 0 \leq y \leq 3; \\ 0, & \text{otherwise.} \end{cases}$$

- a) Find the value of the constant c .
- b) Find the probability $P[A] = P[2 \leq X \leq 3, 1 \leq Y \leq 3]$.
- c) Find the probability $P[B] = P[Y \geq X]$.

$$\int_0^5 \int_0^3 f_{X,Y}(x,y) dy dx = 1$$

$$\Rightarrow \int_0^5 \int_0^3 c dy dx = 1$$

$$\Rightarrow \int_0^5 \left[cy \Big|_0^3 \right] dx = 1$$

$$\Rightarrow \int_0^5 3c dx = 1$$

$$\Rightarrow 3cx \Big|_0^5 = 3c(5 - 0) = 1$$

$$\Rightarrow 15c = 1$$

$$\Rightarrow c = \frac{1}{15}.$$

$$\begin{aligned}
P[A] &= \int_2^3 \int_1^3 \frac{1}{15} dy dx \\
&= \int_2^3 \left[\frac{y}{15} \Big|_1^3 \right] dx \\
&= \int_2^3 \frac{3-1}{15} dx \\
&= \frac{2x}{15} \Big|_2^3 \\
&= \frac{2}{15} (3-2) \\
&= \frac{2}{15}.
\end{aligned}$$

Oppositely, the same probability can be found by calculating the ratio of the desired area to the total area. The area of the desired region is $A = 1 \times 2 = 2$ and the total area is $5 \times 3 = 15$. Therefore, the probability is

$$P[A] = \frac{\text{Area of } A}{\text{Total Area}} = \frac{2}{3}.$$

$$\begin{aligned}
 P[B] &= \int_0^3 \int_x^3 \frac{1}{15} dy dx \\
 &= \int_0^3 \left. \frac{y}{15} \right|_x^3 dx \\
 &= \int_0^3 \frac{3-x}{15} dx \\
 &= \left. \frac{3}{15} x \right|_0^3 - \left. \frac{1}{30} x^2 \right|_0^3 \\
 &= \frac{9}{15} - \frac{9}{30} \\
 &= \frac{9}{15} = \frac{3}{5}.
 \end{aligned}$$

Example 4.7. The JPDF of random variables X and Y is given by

$$P_X(x) = \begin{cases} 2, & \text{for } 0 \leq y \leq x \leq 1; \\ 0, & \text{otherwise.} \end{cases}$$

Find the joint CDF of X and Y .

Case 1: Region R_1

For region R_1 , either $x < 0$ or $y < 0$, and the intersection of the areas R_1 and R produces an empty set. Therefore,

$$F_{X,Y}(x,y) = 0 \quad \text{if } x < 0 \text{ or } y < 0. \quad (4.23)$$

2. Case 2: Region R_2

For region R_2 , $x > 1$ and $0 \leq y \leq 1$. The intersection of the R_2 and R produces an area as shown in the shaded region of Fig. -.-.

$$\begin{aligned}
 F_{X,Y}(x,y) &= \int_0^y \int_y^1 2dx dy \\
 &= \int_0^y \left[2x \Big|_y^1 \right] dy \\
 &= \int_0^y 2(1-y) dy \\
 &= \int_0^y 2dy - \int_0^y 2y dy \\
 &= 2y \Big|_0^y - \frac{2y^2}{2} \Big|_0^y \\
 &= 2y - y^2.
 \end{aligned}$$

3. Case 3: Region R_3

For region R_3 , $0 \leq x \leq 1$ and $y > x$.

$$\begin{aligned}
 F_{X,Y}(x,y) &= \int_0^x \int_y^x 2dx dy \\
 &= \int_0^x \left[2x \Big|_y^x \right] dy \\
 &= \int_0^x (2x - 2y) dy \\
 &= 2xy \Big|_0^x - \frac{2y^2}{2} \Big|_0^x \\
 &= 2x^2 - x^2 \\
 &= x^2.
 \end{aligned}$$

4. **Case 4: Region R_4**

For region R_4 , $0 \leq y \leq x \leq 1$.

$$\begin{aligned}
 F_{X,Y}(x,y) &= \int_0^y \int_y^x 2dx dy \\
 &= \int_0^y \left[2x \Big|_y^x \right] dy \\
 &= \int_0^y (2x - 2y) dy \\
 &= 2xy \Big|_0^y - \frac{2y^2}{2} \Big|_0^y \\
 &= 2xy - y^2.
 \end{aligned}$$

5. Case 5: Region R_5

For region R_5 , $x > 1$ and $y > 1$.

$$F_{X,Y}(x, y) = 1, \quad \text{if } x > 1 \text{ and } y > 1.$$

$$F_{X,Y}(x,y) = \begin{cases} 0, & \text{for } x < 0 \text{ or } y < 0; \\ 2y - y^2, & \text{for } x > 1, 0 \leq y \leq 1; \\ x^2, & \text{for } 0 \leq x \leq 1, y \geq x; \\ 2xy - y^2, & \text{for } 0 \leq y \leq x \leq 1; \\ 1, & \text{for } x > 1, y > 1. \end{cases}$$

Marginal probability density function

Definition 4.9. *Marginal Probability Density Function.* If X and Y are random variable with joint PDF $f_{X,Y}(x,y)$, then the marginal PDF of X and Y are given by

$$f_X(x) = \int_{-\infty}^{\infty} f_{x,y}(x,y)dy, \quad (4.28)$$

and

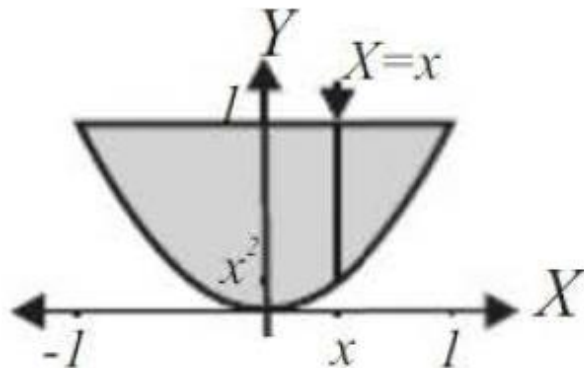
$$f_Y(y) = \int_{-\infty}^{\infty} f_{x,y}(x,y)dx. \quad (4.29)$$

Example 4.8. The joint PDF of X and Y is given by

$$f_{X,Y}(x,y) = \begin{cases} \frac{5}{4}y, & \text{for } -1 \leq x \leq 1, x^2 \leq y \leq 1; \\ 0, & \text{otherwise.} \end{cases}$$

Find the marginal PDFs of X and Y .

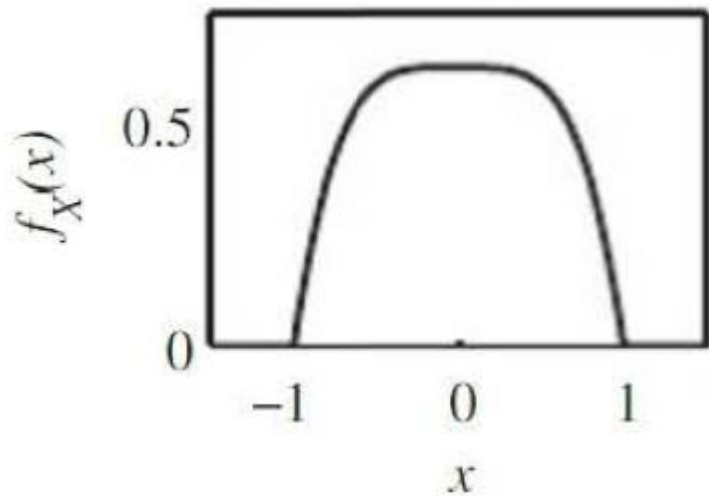
Solution. When $x < -1$ or $x > 1$, then $f_{X,Y}(x,y) = 0$, and therefore, $f_X(x) = 0$. Therefore, the marginal PDF of X is defined for the interval $-1 \leq x \leq 1$. The marginal



$$\begin{aligned}
 f_X(x) &= \int_{x^2}^1 f_{X,Y}(x,y)dy \\
 &= \int_{x^2}^1 \frac{5}{4}ydy \\
 &= \frac{5}{8}y^2 \Big|_{x^2}^1 \\
 &= \frac{5}{8}(1 - x^4).
 \end{aligned}$$

The complete expression for marginal PDF of X is

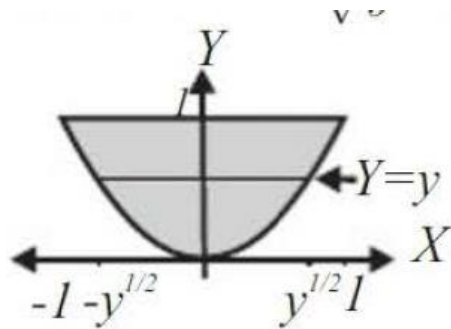
$$f_X(x) = \begin{cases} \frac{5}{8}(1 - x^4), & \text{for } -1 \leq x \leq 1; \\ 0, & \text{otherwise.} \end{cases}$$



Now, we calculate the marginal PDF of Y for the value y , by integrating the joint PDF of X and Y , for all possible values of x with the following limits and are shown in Fig. -.-:

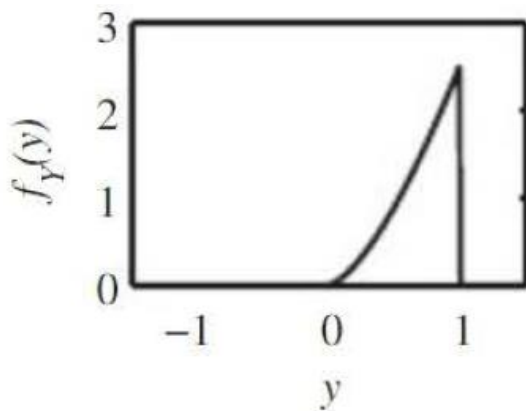
$$\begin{aligned}x^2 &= y \\ \Rightarrow x &= \pm\sqrt{y}.\end{aligned}$$

$$\begin{aligned}
 f_Y(y) &= \int_{-\sqrt{y}}^{+\sqrt{y}} \frac{5}{4} y dx \\
 &= \frac{5}{4} y x \Big|_{-\sqrt{y}}^{+\sqrt{y}} \\
 &= \frac{5}{4} y (\sqrt{y} + \sqrt{y}) \\
 &= \frac{5}{2} \cdot y \cdot y^{\frac{1}{2}} \\
 &= \frac{5}{2} \cdot y^{\frac{3}{2}}.
 \end{aligned}$$



The complete expression of the marginal PDF of Y is given by

$$f_Y(y) = \begin{cases} \frac{5}{2}y^{\frac{3}{2}}, & \text{for } 0 \leq y \leq 1; \\ 0, & \text{otherwise.} \end{cases}$$



Example 4.9. The joint PDF of random variables X and Y is given by

$$f_{X,Y}(x,y) = \begin{cases} \lambda xy^2, & \text{for } 0 \leq x \leq y \leq 1; \\ 0, & \text{otherwise.} \end{cases}$$

1. Determine the value of λ .
2. Find the marginal probability density function of X and Y .
3. Find $E[X]$ and $E[Y]$.

Expected value of a Function of two random variables

For random variables X and Y , the expected value of $W = g(X, Y)$ is

$$\text{Discrete: } E[W] = \sum_{x \in S_X} \sum_{y \in S_Y} g(x, y) P_{X,Y}(x, y);$$

$$\text{Continuous: } E[W] = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} g(x, y) f_{X,Y}(x, y) \, dx dy.$$

$$E[a_1 g_1(X, Y) + \cdots + a_n g_n(X, Y)] = a_1 E[g_1(X, Y)] + \cdots + a_n E[g_n(X, Y)]$$

For any two random variables

$$E[X + Y] = E[X] + E[Y]$$

Covariance and correlation

- The variance of a RV X , $V[X] = E[(X - \mu_X)^2]$, represents the average fluctuation of the values of the RV from its expectation μ_X
- It measures the dispersion or spread of the distribution of X about its mean
- For two jointly distributed RVs X and Y , the $V[X]$ and $V[Y]$ determine the individual dispersions or spreads of X and Y independently
 - $V[X]$ measures the dispersion along the x -axis
 - $V[Y]$ measures the dispersion along the y -axis
 - The joint dispersion is missing
- The joint dispersion or spread of X and Y can be measured along the direction $(ax + by)$ by the variance $V[aX + bY]$

$$\begin{aligned}
\text{Var}(aX + bY) &= E[(aX + bY)E(aX + bY)]^2 \\
&= E[(aX + bY)aE(X)bE(Y)]^2 \\
&= E\left[a(X - E[X]) + b(Y - E[Y])\right]^2 \\
&= E\left[a^2(X - E[X])^2 + b^2(Y - E[Y])^2 + 2ab(X - E[X])(Y - E[Y])\right]^2 \\
&= a^2\text{Var}[X] + b^2\text{Var}[Y] + 2abE\left[(X - E[X])(Y - E[Y])\right].
\end{aligned}$$

The joint dispersion or spread of X and Y can be measured in any direction ($ax + by$) if we know

- Variance of X , $V[X]$ Determine the dispersion of X and Y
- Variance of Y , $V[Y]$ independently
- $E[(X - \mu_X)(Y - \mu_Y)]$ Should give the measure of the joint dispersion

Definition 4.10. *Covariance.* Let X and Y be random variables defined over the same probability space and such that $E[X] < \infty$ and $E[Y] < \infty$. The covariance between X and Y is defined by:

$$\text{Cov}[X, Y] = E\left[(X - E[X])(Y - E[Y])\right]. \quad (4.39)$$

Note that $\text{Cov}[X, X] = \text{Var}[X]$.

Covariance random variables X and Y is also denoted by C_{XY} .

Let a and b be real numbers; for random variables X and Y ,

$$\text{Var}[aX + bY] = a^2\text{Var}[X] + b^2\text{Var}[Y] + 2ab\text{Cov}[X, Y].$$

In particular, if $a = 1$ and $b = 1$, this gives

$$\text{Var}[X + Y] = a^2\text{Var}[X] + b^2\text{Var}[Y] + 2\text{Cov}[X, Y].$$

Similarly, if $a = 1$ and $b = -1$, it gives

$$\text{Var}[X - Y] = a^2\text{Var}[X] + b^2\text{Var}[Y] - 2\text{Cov}[X, Y].$$

Letting $\mu_X = E[X]$, and $\mu_Y = E[Y]$, an alternative formula for $\text{Cov}[X, Y] = E\left[(X - E[X])(Y - E[Y])\right]$ is calculated by the expansion of $E\left[(X - E[X])(Y - E[Y])\right]$:

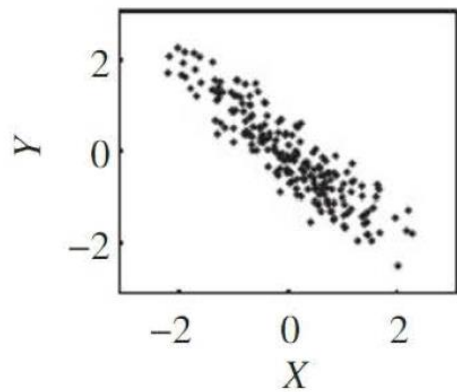
$$\begin{aligned}\text{Cov}[X, Y] &= E\left[(X - \mu_X)(Y - \mu_Y)\right] \\ &= E\left[XY - \mu_X Y - \mu_Y X + \mu_X \mu_Y\right] \\ &= E[XY] - \mu_X E[Y] - \mu_Y E[X] + \mu_X \mu_Y \\ &= E[XY] - \mu_X \mu_Y - \mu_Y \mu_X + \mu_X \mu_Y \\ &= E[XY] - \mu_X \mu_Y \\ &= E[XY] - E[X]E[Y]\end{aligned}$$

For random variables X and Y , the C_{XY} might be

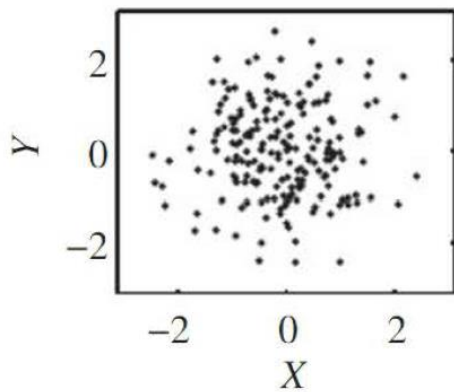
- Positive: X and Y are positively correlated – increase or decrease together
- Negative: X and Y are negatively correlated – X increases while Y decreases, or vice versa
- Zero: X and Y are uncorrelated

Example:

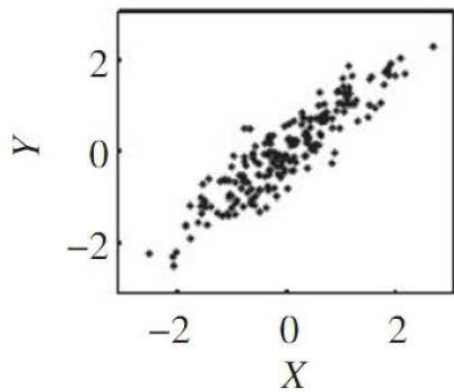
- The Study-time of a student and his CGPA
- The distance of a student from white board and his CGPA



(a) $\rho_{X,Y} = -0.9$



(b) $\rho_{X,Y} = 0$



(c) $\rho_{X,Y} = 0.9$

Example: Suppose we perform an experiment in which we measure X and Y in centimeters (e.g., the height of two students). However, if we change units and measure the height in meters, we will perform the same experiment except we observe $\hat{X} = X/100$ and $\hat{Y} = Y/100$. In this case $\hat{X}$ and $\hat{Y}$ have covariance

$$\begin{aligned}\text{Cov}[\hat{X}\hat{Y}] &= E[(\hat{X} - \mu_{\hat{X}})(\hat{Y} - \mu_{\hat{Y}})] \\ &= \frac{E[(X - \mu_X)(Y - \mu_Y)]}{10000} \\ &= \frac{\text{Cov}[X, Y]}{10000} \text{m}^2.\end{aligned}$$

Changing the unit of measurement from cm^2 to m^2 reduces the covariance by a factor of 10,000.

Correlation Coefficient

A normalized version of C_{XY} , that indicates the relationship of two RVs variables regardless of the measurement units, is called the correlation coefficient and is denoted by ρ_{XY} .

$$\rho_{X,Y} = \frac{\text{Cov}[X, Y]}{\sqrt{\text{Var}[X]\text{Var}[Y]}} = \frac{\text{Cov}[X, Y]}{\sigma_X \sigma_Y}$$

Properties of Covariance:

- If X and Y are independent RVs with $0 < \sigma_X^2 < \infty$ and $0 < \sigma_Y^2 < \infty$, then
$$C_{XY} = \rho_{XY} = 0$$

However, the converse of the above property may not be true.

Example 4.11. Let X be uniformly distributed over $(-1, 1)$ and $Y = X^2$. Then

$$\text{Cov}[X, Y] = E[X^3] - E[X]E[X^2] = 0.$$

since $E[X] = 0$ and $E[X^3] = 0$. Thus the perfectly related random variables X and Y are uncorrelated.

- If $X_1, \dots, X_n$ are RVs such that $V[X_i] < \infty$, for $i = 1, 2, \dots, n$

$$\text{Var}\left(\sum_{i=1}^n X_i\right) = \sum_{i=1}^n \text{Var}(X_i) + 2 \sum_{i < j} \text{Cov}(X_i, X_j).$$

- If $X_1, \dots, X_n$ are uncorrelated RVs

$$\text{Var}\left(\sum_{i=1}^n X_i\right) = \sum_{i=1}^n \text{Var}(X_i).$$

- If $X_1, \dots, X_n$ are RVs

$$\text{Var}\left(\sum_{i=1}^n a_i X_i\right) = \sum_{i=1}^n a_i^2 \text{Var}(X_i) + 2 \sum_{i < j} a_i a_j \text{Cov}(X_i, X_j)$$

In particular, for $a_i = 1, 1 \leq i \leq n$

$$\text{Var}\left(\sum_{i=1}^n X_i\right) = \sum_{i=1}^n \text{Var}(X_i) + 2 \sum_{i < j} \text{Cov}(X_i, X_j).$$

Example 4.12. Let X be the lifetime of an electronic system and Y be the lifetime of one of its components. Suppose that the electronic system fails if the component does (but not necessarily vice versa). Furthermore, suppose that the joint probability density function of X and Y (in years) is given by

$$f_{X,Y}(x,y) = \begin{cases} \frac{1}{49}e^{-\frac{y}{7}}, & \text{for } 0 \leq x \leq y \leq \infty; \\ 0, & \text{otherwise.} \end{cases}$$

- (1) Determine the expected value of the remaining lifetime of the component when the system dies.
- (2) Find the covariance of X and Y .

(1) The remaining lifetime of the component when the system dies is YX . So desired quantity is

$$\begin{aligned} E[Y - X] &= \int_0^\infty \int_0^y (y - x) \frac{1}{49} e^{-\frac{y}{7}} dx dy \\ &= \frac{1}{49} \int_0^\infty e^{-\frac{y}{7}} \left(y^2 - \frac{y^2}{2} \right) dy \\ &= \frac{1}{98} \int_0^\infty y^2 e^{-\frac{y}{7}} dy \\ &= 7. \end{aligned}$$

(2) To find $\text{Cov}[X, Y] = E[XY] - E[X]E[Y]$, note that

$$\begin{aligned} E[XY] &= \int_0^\infty \int_0^y (xy) \frac{1}{49} e^{-\frac{y}{7}} dx dy \\ &= \frac{1}{49} \int_0^\infty y e^{-\frac{y}{7}} \left(\int_0^y x dx \right) dy \\ &= \frac{1}{98} \int_0^\infty y^3 e^{-\frac{y}{7}} dy \\ &= \frac{14406}{98} \\ &= 147. \end{aligned}$$

$$E[X] = \int_0^\infty \int_0^y x \frac{1}{49} e^{-\frac{y}{7}} dx dy = 7,$$

$$E[Y] = \int_0^\infty \int_0^y y \frac{1}{49} e^{-\frac{y}{7}} dx dy = 14.$$

Therefore, $\text{Cov}[X, Y] = 1477(14) = 49$. Note that $\text{Cov}[X, Y] > 0$ is expected because X and Y are positively correlated.

Definition 5.7 Correlation

The *correlation* of X and Y is $r_{X,Y} = E[XY]$

(a) $\text{Cov}[X, Y] = r_{X,Y} - \mu_X \mu_Y.$

(b) $\text{Var}[X + Y] = \text{Var}[X] + \text{Var}[Y] + 2 \text{Cov}[X, Y].$

(c) If $X = Y$, $\text{Cov}[X, Y] = \text{Var}[X] = \text{Var}[Y]$ and $r_{X,Y} = E[X^2] = E[Y^2].$

Example 5.17

For the integrated circuits tests in Example 5.3, we found in Example 5.5 that the probability model for X and Y is given by the following matrix.

$P_{X,Y}(x,y)$	$y = 0$	$y = 1$	$y = 2$	$P_X(x)$
$x = 0$	0.01	0	0	0.01
$x = 1$	0.09	0.09	0	0.18
$x = 2$	0	0	0.81	0.81
$P_Y(y)$	0.10	0.09	0.81	

Find $r_{X,Y}$ and $\text{Cov}[X,Y]$.

$$\begin{aligned}
 r_{X,Y} = \text{E}[XY] &= \sum_{x=0}^2 \sum_{y=0}^2 xyP_{X,Y}(x,y) \\
 &= (1)(1)0.09 + (2)(2)0.81 = 3.33.
 \end{aligned}$$

$$\text{E}[X] = (1)(0.18) + (2)(0.81) = 1.80,$$

$$\text{E}[Y] = (1)(0.09) + (2)(0.81) = 1.71.$$

$$\text{Cov}[X, Y] = 3.33 - (1.80)(1.71) = 0.252.$$

Definition 5.8 — Orthogonal Random Variables

Random variables X and Y are *orthogonal* if $r_{X,Y} = 0$.

Definition 5.9 — Uncorrelated Random Variables

Random variables X and Y are *uncorrelated* if $\text{Cov}[X, Y] = 0$.

Multivariate Random Variables

Random Experiments with more than two RVs

Outcomes of the experiments are represented by points on an n dimensional space

The probability models associated are

1. Multivariate joint PMF (MJPMF) – represents the probability of occurring an outcome $x_1, \dots, x_n$ – simultaneously n RVs have the values (as real numbers)
$$X_1 = x_1, \dots, X_n = x_n$$
2. Multivariate Joint PDF (MJPDF) – represents the probability density at a point $(x_1, \dots, x_n)$ in the n -dimensional space
3. Multivariate Joint CDF (MJCDF) – represents the total probability of an n -dimensional space associated with the point $(x_1, \dots, x_n)$ and defined by the lines $X_1 = x_1, \dots, X_n = x_n$.
4. Marginal PMFs/JPMFs/MJPMFs, Marginal PDFs/JPDFs/MJPDFs and marginal CDFs/JCDFs/MJCDFs of all the subsets of RVs

Multivariate Discrete Random Variables

Multivariate Joint Probability Mass Function (MJPMF)

$$P_{X_1, \dots, X_n}(x_1, \dots, x_n) = P[X_1 = x_1, \dots, X_n = x_n].$$

1. $P_{X_1, \dots, X_n}(x_1, \dots, x_n) \geq 0$.
2. If for some i , $1 \leq i \leq n$, $x_i \notin S_{X_i}$, then

$$P_{X_1, \dots, X_n}(x_1, \dots, x_n) = 0.$$

3. $\sum_{x_i \in S_{X_i}, 1 \leq i \leq n} P_{X_1, \dots, X_n}(x_1, \dots, x_n) = 1$.

Marginal PMF for Multivariate distribution

$$\begin{aligned}P_{X_i}(x_i) &= P[X_i = x_i] \\&= P[X_i = x_i, x_j \in S_{X_j}], \quad i \leq j \leq n, i \neq j \\&= \sum_{x_1 \in S_{X_1}} \cdots \sum_{x_{i-1} \in S_{X_{i-1}}} \sum_{x_{i+1} \in S_{X_{i+1}}} \cdots \sum_{x_n \in S_{X_n}} P_{X_1, \dots, X_n}(x_1, \dots, x_n), \quad 1 \leq i \leq n\end{aligned}$$

$$\begin{aligned}P_{X_i, X_j}(x_i, x_j) &= P[X_i = x_i, X_j = x_j] \\&= P[X_i = x_i, X_j = x_j, x_k \in S_{X_k}], \quad i \leq k \leq n, k \neq i, j \\&= \sum_{x_1 \in S_{X_1}} \cdots \sum_{x_{i-1} \in S_{X_{i-1}}} \sum_{x_{i+1} \in S_{X_{i+1}}} \cdots \sum_{x_{j-1} \in S_{X_{j-1}}} \sum_{x_{j+1} \in S_{X_{j+1}}} \cdots \sum_{x_n \in S_{X_n}} P_{X_1, \dots, X_n}(x_1, \dots, x_n)\end{aligned}$$

Marginal multivariate joint PMF

$$P_{X_{j_1}, \dots, X_{j_k}}(x_1, \dots, x_k) = P[X_{j_1} = x_1, \dots, X_{j_k} = x_k]$$

Multivariate joint cumulative distribution function (MJCDF)

$$\begin{aligned} F_{X_1, \dots, X_n}(x_1, \dots, x_n) &= P[X_1 \leq x_1, \dots, X_n \leq x_n], \quad x_i < +\infty, 1 \leq i \leq n \\ &= \sum_{x_1 \in S_{X_1}} \sum_{x_2 \in S_{X_2}} \dots \sum_{x_n \in S_{X_n}} P_{X_1, \dots, X_n}(x_1, \dots, x_n) \end{aligned}$$

1. $F_{X_1, \dots, X_n}(x_1, \dots, x_n)$ is nondecreasing in each argument.
2. $F_{X_1, \dots, X_n}(x_1, \dots, x_n)$ is right continuous in each argument.
3. $F_{X_1, \dots, X_n}(x_1, \dots, x_n) = 0$, if for at least one random variable $x_i = -\infty$.
4. $F_{X_1, \dots, X_n}(\infty, \dots, \infty) = 1$

Marginal CDF multiple RVs

$$\begin{aligned}F_{X_i}(x) &= P[X_i \leq x] \\&= P[X_1 < \infty, \dots, X_{i-1} < \infty, X_i = x, X_{i+1} < \infty, X_n < \infty] \\&= \sum_{x_1 < \infty} \cdots \sum_{x_{i-1} < \infty} \sum_{x_i < x} \sum_{x_{i+1} < \infty} \cdots \sum_{x_n < \infty} P_{X_1, \dots, X_n}(x_1, \dots, x_n) \\&= \lim_{\substack{x_j \rightarrow \infty \\ 1 \leq j \leq n, j \neq i}} F_{X_1, \dots, X_n}(x_1, \dots, x_n).\end{aligned}$$

Example: In a box, there are 23 balls of which there are 5 red balls, 8 blue balls and 10 green balls. Suppose 7 balls are picked without replacement. Let X, Y and Z represent number of red, blue and green balls. Find the multivariate joint PMF and marginal PMFs of X, Y and Z .

Let the MJPMF is $P_{X,Y,Z}(x, y, z)$.

The probability is the ratio of following two:

1. Number of ways x, y and z balls can be picked which is the product of the following three terms
 - a. Number of ways x balls are picked from 5 red balls
 - b. Number of ways y balls are picked from 8 blue balls
 - c. Number of ways z balls are picked from 10 green balls
2. Number of ways 7 balls are randomly picked from 23 balls

Therefore, the MJPMF of X, Y and Z $P_{X,Y,Z}(x, y, z)$, for $0 \leq x \leq 5, 0 \leq y \leq 8, 0 \leq z \leq 10$, and $x + y + z = 7$, can be given by

$$P_{X,Y,Z}(x, y, z) = \frac{\binom{5}{x} \binom{8}{y} \binom{10}{z}}{\binom{23}{7}}$$

and $P_{X,Y,Z}(x, y, z) = 0$ otherwise.

The marginal PMF of X , $P_X(x)$, can be obtained in the following way:

$$\begin{aligned}
 P_X(x) &= P[X = x] \\
 &= \sum_{\substack{x+y+z=7 \\ 0 \leq y \leq 7 \\ 0 \leq z \leq 7}} P_{X,Y,Z}(x, y, z) \\
 &= \sum_{y=0}^{7-x} \frac{\binom{5}{x} \binom{8}{y} \binom{10}{7-x-y}}{\binom{23}{7}} \\
 &= \frac{\binom{5}{x}}{\binom{23}{7}} \sum_{y=0}^{7-x} \binom{8}{y} \binom{10}{7-x-y}.
 \end{aligned}$$

$$\sum_{y=0}^{7-x} \binom{8}{y} \binom{10}{7-x-y}$$

Is the total number of ways we can choose $(7 - x)$ balls of blue or green colors from 18 balls

Which is, number of ways $(7 - x)$ balls are picked from 18 balls and is $\binom{18}{7-x}$

Therefore, the marginal PMF of X is

$$P_X(x) = \frac{\binom{5}{x} \binom{18}{7-x}}{\binom{23}{7}}.$$

Similarly,

$$P_Y(y) = \frac{\binom{8}{y} \binom{18}{7-y}}{\binom{23}{7}}$$

$$P_Z(z) = \frac{\binom{10}{z} \binom{18}{7-z}}{\binom{23}{7}}.$$

Well-Known Multivariate RVs

Definition 5.1. (*Multivariate hypergeometric distribution*) Suppose an urn contains items of r different types, where the number of the i -th type is n_i and the number of total items is $n_1 + n_2 + \cdots + n_r$. Now, n items are drawn from the urn at random. Let the random variable X_i , where $i = 1, 2, \dots, r$, represent the number of items drawn of the i -th type. Then, the multivariate joint probability mass function of $X_1, X_2, \dots, X_r$ is given by

$$P_{X_1, \dots, X_r}(x_1, \dots, x_r) = \frac{\binom{n_1}{x_1} \binom{n_2}{x_2} \cdots \binom{n_r}{x_r}}{\binom{n_1 + n_2 + \cdots + n_r}{n}},$$

where $x_1 + x_2 + \cdots + x_r = n$. The above MJPMF is known as the multivariate hypergeometric distribution.

Multinomial Distributions. A discrete random vector $\mathbf{X} = (X_1, \dots, X_k)$ whose p.f. is given by Eq. (5.9.1) has the *multinomial distribution with parameters n and $\mathbf{p} = (p_1, \dots, p_k)$* .

$$\begin{aligned} f(\mathbf{x}|n, \mathbf{p}) &= \Pr(\mathbf{X} = \mathbf{x}) = \Pr(X_1 = x_1, \dots, X_k = x_k) \\ &= \begin{cases} \binom{n}{x_1, \dots, x_k} p_1^{x_1} \cdots p_k^{x_k} & \text{if } x_1 + \cdots + x_k = n, \\ 0 & \text{otherwise.} \end{cases} \end{aligned}$$

$$\binom{n}{x_1, \dots, x_k} = \frac{n!}{x_1! x_2! \cdots x_k!}.$$

Example

Attendance at a Baseball Game. Suppose that 23 percent of the people attending a certain baseball game live within 10 miles of the stadium, 59 percent live between 10 and 50 miles from the stadium, and 18 percent live more than 50 miles from the stadium. Suppose also that 20 people are selected at random from the crowd attending the game. We shall determine the probability that seven of the people selected live within 10 miles of the stadium, eight of them live between 10 and 50 miles from the stadium, and five of them live more than 50 miles from the stadium.

$$\frac{20!}{7!8!5!}(0.23)^7(0.59)^8(0.18)^5 = 0.0094.$$

Multivariate Continuous Random Variables

Multivariate Joint CDF

$$F_{X_1, \dots, X_n}(x_1, \dots, x_n) = P[X_1 \leq x_1, \dots, X_n \leq x_n]$$

Multivariate joint PDF

If the joint distribution of n (where $n > 2$) random variables $X_1, \dots, X_n$ is continuous, then the multivariate joint PDF, $f_{X_1, \dots, X_n}(x_1, \dots, x_n)$, can be derived from the MJCDF using the following relation

$$f_{X_1, \dots, X_n}(x_1, \dots, x_n) = \frac{\partial^n F_{X_1, \dots, X_n}(x_1, \dots, x_n)}{\partial x_1, \dots, \partial x_n},$$

$$F_{X_1, \dots, X_n}(x_1, \dots, x_n) = \int_{-\infty}^{x_1} \cdots \int_{-\infty}^{x_n} f_{X_1, \dots, X_n}(x_1, \dots, x_n) dx_1 \dots dx_n.$$

Marginal density function for multiple RVs

If the multivariate joint PDF of $X_1, \dots, X_n$ is $f_{X_1, \dots, X_n}(x_1, \dots, x_n)$, then the marginal PDF of X_1 , $f_{X_1}(x_1)$, can be given by

$$f_{X_1}(x_1) = \underbrace{\int_{-\infty}^{+\infty} \cdots \int_{-\infty}^{+\infty}}_{n-1} f_{X_1, \dots, X_n}(x_1, \dots, x_n) dx_2 \dots dx_n.$$

variables. For example, if the multivariate joint PDF of 4 random variables $X_1, \dots, X_4$ $f_{X_1, \dots, X_4}(x_1, \dots, x_4)$, then the marginal joint PDF of X_2 and X_4 , $f_{X_2, X_4}(x_2, x_4)$, can be

$$f_{X_2, X_4}(x_2, x_4) = \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} f_{X_1, \dots, X_4}(x_1, \dots, x_4) dx_1 dx_3.$$

Example 5.2. Suppose that a machine has three parts, and part i will fail at time X_i for $i = 1, 2, 3$. The following function may be the multivariate joint CDF of X_1, X_2 and X_3 .

$$F_{X_1, X_2, X_3}(x_1, x_2, x_3) = \begin{cases} (1 - e^{-x_1})(1 - e^{-2x_2})(1 - e^{-3x_3}), & \text{for } x_1, x_2, x_3 \geq 0 \\ 0, & \text{otherwise.} \end{cases}$$

- a) Find the multivariate joint PDF of X_1, X_2, X_3 .
- b) Find the marginal joint PDF of X_1 and X_2 .

a) The multivariate joint PDF of X_1, X_2, X_3 is given by

$$\begin{aligned} f_{X_1, X_2, X_3}(x_1, x_2, x_3) &= \frac{\partial^3 F_{X_1, X_2, X_3}(x_1, x_2, x_3)}{\partial x_1 \partial x_2 \partial x_3} \\ &= e^{x_1} . 2e^{2x_2} . 3e^{3x_3} \\ &= 6e^{-x_1-2x_2-3x_3}, \quad x_1, x_2, x_3 \geq 0. \end{aligned}$$

b) The marginal joint CDF of x_1 and x_3 is given by

$$\begin{aligned} F_{X_1, X_3}(x_1, x_3) &= \int_{-\infty}^{+\infty} F_{X_1, X_2, X_3}(x_1, x_2, x_3) dx_2 \\ &= \int_{-\infty}^{+\infty} (1 - e^{-x_1})(1 - e^{-2x_2})(1 - e^{-3x_3}) dx_2 \\ &= (1 - e^{-x_1})(1 - e^{-3x_3}) \int_{-\infty}^{+\infty} (1 - e^{-2x_2}) dx_2 \\ &= (1 - e^{-x_1})(1 - e^{-3x_3}) \quad x_1, x_2 \geq 0. \end{aligned}$$

Random Variables $Y_1, \dots, Y_4$ have the joint PDF

$$f_{Y_1, \dots, Y_4}(y_1, \dots, y_4) = \begin{cases} 4 & 0 \leq y_1 \leq y_2 \leq 1, 0 \leq y_3 \leq y_4 \leq 1, \\ 0 & \text{otherwise.} \end{cases}$$

Find the marginal PDFs $f_{Y_1, Y_4}(y_1, y_4)$, $f_{Y_2, Y_3}(y_2, y_3)$ and $f_{Y_3}(y_3)$.

$$f_{Y_1, Y_4}(y_1, y_4) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f_{Y_1, \dots, Y_4}(y_1, \dots, y_4) \, dy_2 \, dy_3.$$

For $0 \leq y_1 \leq 1$ and $0 \leq y_4 \leq 1$

$$f_{Y_1, Y_4}(y_1, y_4) = \int_{y_1}^1 \int_0^{y_4} 4 \, dy_3 \, dy_2 = 4(1 - y_1)y_4.$$

The complete expression for $f_{Y_1, Y_4}(y_1, y_4)$ is

$$f_{Y_1, Y_4}(y_1, y_4) = \begin{cases} 4(1 - y_1)y_4 & 0 \leq y_1 \leq 1, 0 \leq y_4 \leq 1, \\ 0 & \text{otherwise.} \end{cases}$$

Similarly, for $0 \leq y_2 \leq 1$ and $0 \leq y_3 \leq 1$,

$$f_{Y_2, Y_3}(y_2, y_3) = \int_0^{y_2} \int_{y_3}^1 4 dy_4 dy_1 = 4y_2(1 - y_3).$$

The complete expression for $f_{Y_2, Y_3}(y_2, y_3)$ is

$$f_{Y_2, Y_3}(y_2, y_3) = \begin{cases} 4y_2(1 - y_3) & 0 \leq y_2 \leq 1, 0 \leq y_3 \leq 1, \\ 0 & \text{otherwise.} \end{cases}$$

For $0 \leq y_3 \leq 1$

$$f_{Y_3}(y_3) = \int_{-\infty}^{\infty} f_{Y_2, Y_3}(y_2, y_3) dy_2 = \int_0^1 4y_2(1 - y_3) dy_2 = 2(1 - y_3).$$

The complete expression is

$$f_{Y_3}(y_3) = \begin{cases} 2(1 - y_3) & 0 \leq y_3 \leq 1, \\ 0 & \text{otherwise.} \end{cases}$$